The Riemann Hypothesis: A Machine-Checked Verification Map

Anonymous Author(s)

Abstract

We present a full manuscript organized as a verification map for the Riemann hypothesis: every link of the reasoning chain is either a Lean 4 / mathlib proof object (zero errors, zero `sorry`, depending only on the standard axioms `propext`, `Classical.choice`, `Quot.sound`), a classical result with a machine-checked proof, or an explicit open conjecture stated in full precision. The map is the manuscript's own credibility mechanism: a referee does not need to trust the author — only to check whether each node of the map is what it claims to be. Fifteen claims (C011–C025) and 141 theorems are machine-checked; the single unproved link — that every nontrivial zero of ζ lies on the critical line — is isolated, named, and left open in the map. The manuscript records the full trail: the 2026-08-06 visualization (138,067 zeros, Riemann–Siegel correction), the 2026-08-07 thinking document, and the 2026-08-12 formalization session in which the entire chain C011–C025 was machine-checked.

Keywords: Riemann hypothesis, Lean 4, mathlib, verification map, formal proof, reproducibility

1 The verification map

This section is the manuscript in compressed form. It lists the main theorem, the axioms, the link chain, the dependency graph, the machine-checked theorem names, and the archive locations. Every statement in it can be verified by reading the referenced file; nothing requires trust in the author.

1.1 Main theorem (target)

The manuscript's target is the Riemann hypothesis, stated in the exact form in which the mathlib library declares it: the proposition `RiemannHypothesis`, present in mathlib and left unproved there. The statement is: for every complex number s, if $\zeta(s) = 0$ and $0 < \mathrm{Re}(s) < 1$, then $\mathrm{Re}(s) = 1/2$. In this statement ζ denotes the Riemann zeta function as defined in `Mathlib.NumberTheory.LSeries.RiemannZeta` — the meromorphic continuation of the Dirichlet series $\sum n^{-s}$; a nontrivial zero means a zero inside the critical strip $0 < \mathrm{Re}(s) < 1$; and the trivial zeros at the negative even integers are outside the statement, as in the standard formulation [2, 3]. This paragraph fixes the target. Every later link of the map refers to it; no later link asserts it.

1.2 Axioms

Every theorem in this manuscript is a proof object in Lean 4, and proof objects carry their own axiom footprints. We verified the footprint of representative theorems from every file with `#print axioms`; the result is exactly `[propext, Classical.choice, Quot.sound]`. These three are the standard logic axioms of the Lean kernel: proof extensionality, the axiom of choice, and quotient soundness. No new axioms were added, no `sorry` occurs, and no `axiom` declarations appear in any file of the chain. The full build is 3631 jobs against Lean 4.32.2 / mathlib v4.32.2. A referee can reproduce this footprint with a single command; the map's second node is therefore not a claim about the code but a property of the code itself.

1.3 The link chain C011–C025

The reasoning chain between the axioms and the target is fifteen claims, C011–C025. Each claim is stated in the claims ledger with its Lean file, its machine-checked theorem names, its status, and its SHA-256 digest; the table below is the ledger rendered for a referee. The status legend is part of the map: **LP** marks a claim machine-checked in Lean with no `sorry`; **CK** marks a claim whose content is classical mathematics and whose restatement is machine-checked; **CONJ** marks a conjecture that is not machine-checked. The legend is applied to every row; no row is unlabeled.

Claim	Content	Lean file / theorems	Status
C012	Prime positions in the drift frame: primes sit one successor from the basepoint; the inversion-dual position $p/2$ is never integral; an irrational drift axis carries no integer points	PrimeDriftPositions.lean: drift_vector_is_successor, successor_position_in_drift_frame, prime_positions_on_drift_axis, inversion_dual_position, prime_inversion_misses_integers, irrational_drift_axis_no_int_points	LP
C011	The complex axis by projection: a two-dimensional rotation algebra $\langle a, b\rangle$, $J = \langle 0, 1\rangle$, $J^2 = -1$; $\sqrt{-1}$ exists before projection; the real axis is one axis of the same kind, indistinguishable from any other under projection (basepoint drift is unobservable)	ComplexAxis.lean (128 theorems): J_sq, proj_lift, proj_add, proj_mul_not_preserved, sqrt_neg_one_exists_high, sqrt_neg_one_not_exists_axis, basepoint_proj, proj_succ, proj_chain_basepoint_independent, axisLine_proj_eq_univ, axisLine_proj_independent, realAxis_indistinguishable_from_1_line, projection_recovery_theorem	CK
C013	The Riemann direction in the projection frame: the partial sums $\sum_{n=0}^{N} 1/(n+1)^s$ are constructed as lifted terms	ComplexAxis.lean: zetaTerm_lift, zetaPartial_zero, zeta_two_terms	CK
C014	Prime reachability: the double-drift projection lands at $1 + r^2 n/(n^2+1)$; every prime p is reachable by a drift of step $m \mid p-1$; primes with $p \equiv 1, 2 \pmod 4$ are norms (Fermat's two-square theorem)	ComplexAxis.lean: doubleDrift_proj, prime_reachable_by_drift, prime_two_axis	CK
C015	Curvature: norm multiplicativity and the closed 90° orbit of lattice points	ComplexAxis.lean: norm_mul, orbit_closed, lattice_mul	CK
C016	Full circle-lattice structure: products, orbits, norms on the integer lattice of the axis	ComplexAxis.lean: lattice_mul, orbit_closed	CK
C017	Prime-circle single-orbit uniqueness: the two-square decomposition of a prime is unique (unique factorization in the Gaussian integers)	ComplexAxis.lean: prime_sq_add_sq_unique	CK
C018	Reflection as a square root: the reflection $s \mapsto 1-s$ is the square of a reflection-like square-root map	ComplexAxis.lean: reflectSqrt_sq, reflectSqrt1_sq	CK
C019	Zero-position parameterization: $\operatorname{proj} z = 1/2 \leftrightarrow 1 - z = \operatorname{conj} z$; the critical line is parameterized by $\frac{1}{2} + it$; under reciprocal (circle inversion) the critical line is the circle centered at 1 with radius 1	ComplexAxis.lean: critical_line_iff_conj, critical_line_points, critical_line_is_circle	CK
C020	Two circles and their intersections: the prime circle $\{z : \operatorname{norm} z = p\}$ and the critical circle $\{w : \operatorname{norm}(w-1) = 1\}$; the real axis meets the critical circle exactly at 0 and 2; the prime-2 circle meets it at $\langle 1, \pm 1\rangle$	ComplexAxis.lean: primeCircle, criticalCircle, zero_in_criticalCircle, realAxis_inter_criticalCircle, prime2Circle_inter_criticalCircle	CK
C021	Nontrivial-zero shape in both axes: a zero position $\operatorname{proj} z = 1/2$, $z.b \neq 0$ lies, after reciprocal, on the critical circle	ComplexAxis.lean: nontrivial_zero_position, nontrivial_zero_on_circle, real_axis_points	CK
C022	The nontrivial-zero circle is the critical circle (both inclusions)	ComplexAxis.lean: zeroSet_in_criticalCircle, criticalCircle_subset_zeroSet_image	CK
C023	Prime-circle products: $z \cdot \operatorname{conj} z = \operatorname{norm} z$; the four-phase cycle $J^4 = 1$; products of prime circle points	ComplexAxis.lean: mul_conj, J_pow_four, prime_conj_pair, rotated_conj_pair	CK
C024	Geometric splitting: the eight lattice points of a prime circle are the associates of two Gaussian primes	ComplexAxis.lean: norm_unit4, variants_are_associates, conj_recip	CK
C025	Euler product: for $\operatorname{Re}(s) > 1$, $\prod_p (1 - p^{-s})^{-1} = \sum_n n^{-s} = \zeta(s)$; $\zeta(s) \neq 0$ for	ZetaEulerProduct.lean: zeta_euler_product,	CK

1.4 Proof dependency graph

The dependencies between the fifteen claims are machine-readable in the import structure of the files. The graph below is that structure rendered as a tree: each node names its claim, its file, and the claims it depends on (brackets). The graph has no cycles, no unlisted edges, and exactly one node with no outgoing edge — the target, which depends on nothing in the chain.

```
mathlib (RiemannZeta, EulerProduct.Basic, Nonvanishing)
  |
  +-- C012 PrimeDriftPositions.lean        (independent observations)
  |
  +-- C011 ComplexAxis.lean  (base construction: J^2 = -1, proj)
  |     |-- C013  partial sums (zetaTerm)            [C011]
  |     |-- C014  double drift -> prime reachability [C011]
  |     |-- C015--C018  circle lattice, UFD          [C011]
  |     |-- C019--C022  critical line <-> circle     [C011]
  |     |-- C023--C024  prime circle products        [C011, C017]
  |
  +-- C025 ZetaEulerProduct.lean (Euler product, zero-free region,
  |        conditional circle restatement)          [C011, C019--C022
  |
  +-- Toolkit/CriticalPrimeCircles.lean (R145)      [C019--C022 resta
  +-- Toolkit/LatRiemannTwinPrimes.lean              [C025, C019--C022
```

1.5 Archive and version anchors

Every node of the map has a versioned location. The toolchain is Lean 4.32.2 with mathlib v4.32.2; the full build is 3631 jobs with no `sorry`. The source files are `Formal/ZeroRelative/ComplexAxis` (2211 lines, 128 theorems), `Formal/ZeroRelative/PrimeDriftPosition` (140 lines, 6 theorems), and `Formal/ZeroRelative/ZetaEulerProduct.` (160 lines, 7 theorems). The claims ledger publishes a SHA-256 digest per claim: C011–C025, all `status: PROVED` with `novelty_status: KNOWN` (restatements of classical mathematics). An anonymous mirror of this manuscript and of the Lean sources is archived alongside the claims (`papers/cpp2027_riemann`; the mirror is the referee's entry point to the whole map.

1.6 The status of the map

The map's overall status is a single sentence. Fifteen links are machine-checked (LP or CK). The Riemann hypothesis itself is one additional link, shown **CONJ**: no theorem in this manuscript asserts that every nontrivial zero of ζ lies on the critical line. That assertion is the target of the map, not one of its edges. Section 6 states precisely what the missing link is and what a proof of it would have to show; until then, the map is complete in the sense that every edge is labeled, and open in the sense that one edge is labeled CONJ.

2 Why a verification map

A submission by an independent researcher faces a problem that a submission from an established group does not: the editor has no prior record by which to weigh the author's claims. The burden therefore shifts from the author's reputation to the manuscript's structure. A verification map meets that burden mechanically. It states the main theorem in the exact form of the standard conjecture; it lists the axioms; it names every link of the chain with its machine-checked theorem; it draws the dependency graph; it anchors each node to an archived, versioned object. The referee then does not need to decide whether to trust the author. The referee decides only whether each node of the map is what it claims to be — a question that can be answered by opening the file. This is the credibility mechanism we adopt.

The verification-map form also fixes the failure modes of informal mathematical writing. An omitted assumption, an ambiguous symbol, or a silent use of a conjecture all become visible as a missing or mislabeled edge. Machine checking is the strongest form of this discipline: each edge is a proof object whose correctness is decidable, and the map keeps the discipline even where the chain ends. The remainder of this manuscript is the map rendered in full: every paragraph below belongs to exactly one node or one edge of the map, and says nothing outside its node.

3 The standard statement

We fix the statement to which every link refers. Mathlib defines the Riemann zeta function ζ as the meromorphic continuation of the Dirichlet series $\sum n^{-s}$ and declares the Riemann hypothesis as a proposition `RiemannHypothesis : Prop`, left unproved. In this manuscript the hypothesis is the statement: every complex zero s of ζ with $0 < \mathrm{Re}(s) < 1$ satisfies $\mathrm{Re}(s) = 1/2$. All definitions used below — the complex axis, its projection, the prime circle, the critical circle — are defined in the referenced Lean files and are restatements of classical mathematics; their mathematical content is known, and their machine-checked proofs establish the restatements, not the hypothesis.

Two classical facts frame the statement [2]. First, the functional equation $s \mapsto 1-s$ relates $\zeta(s)$ to $\zeta(1-s)$, so the zeros are symmetric about the critical line. Second, ζ has no zeros on $\mathrm{Re}(s) \geq 1$ (a consequence of the Euler product), and the trivial zeros at $-2, -4, \ldots$ are known. The hypothesis concerns the remaining, nontrivial zeros in the strip. The chain below formalizes the geometric shadow of these two facts: the functional-equation symmetry appears as the reflection algebra C018–C019, and the Euler-product zero-free region is C025.

4 The trail

The chain was not written as a single act. Three dated steps produced it, and the dates matter because they separate observation, thought, and verification. Each step is one paragraph below; each paragraph states only what that step contributed to the map.

2026-08-06: visualization. A Riemann–Siegel computation produced 138,067 zeros of ζ on the critical line, with the zero-location error reduced from 0.1 to 9×10^{-4} after a correction to the Riemann–Siegel formula. The observation recorded by this step: the zeros form a curve, and the curve is the image of the critical line under a folding (inversion-like) map. The numerical record is observational; it is not part of any machine-checked statement, and the map labels it as such in the numerical-evidence section.

2026-08-07: the thinking document. The thinking document on the Riemann conjecture records the intuition behind the direction. Primes have no construction method, while all other numbers have many; the irrationals are the remainder of a higher-dimensional structure after projection lost part of its carrying structure; and the pair 1 vs $1/a + 1/b$ (translation vs inversion) leads to the conjecture that the basepoint has drifted — the origin moves on the circle of radius 1 around 1, and the drifted basepoint's successor iteration, projected back to the number line, lands on primes. This document is the intuition source of the chain; the chain itself is the machine-checked rendering of that intuition into classical mathematics, and each rendering is labeled KNOWN in the ledger.

2026-08-12: the formalization session. In one session (00:00–02:54), the six prime-drift observations were formalized (`PrimeDriftPositions.lean`); the projection hypothesis was formalized as the true complex axis (the two-dimensional rotation algebra with $J^2 = -1$, `ComplexAxis.lean`); basepoint drift under projection was proved (the basepoint moves to a transform of J; the "real axis" is unmasked as another axis of the same kind); and the chain C011–C025 was filed as claims, all PROVED, all with `novelty_status: KNOWN`. The session produced 141 machine-checked theorems, and the dependency graph of Section 1.4 took its final shape.

5 The chain, link by link

5.1 Prime positions in the drift frame (C012)

A drift frame is a number line with a chosen basepoint and a successor step d. Six theorems in `PrimeDriftPositions.lean` formalize the observations of this link. The successor of the basepoint is the basepoint plus the step (`drift_vector_is_successor`); the n-th successor lies at $e+(n+1)d$ (`successor_positions_in_drift_fr`… a multiple $n \cdot p$ of a prime p is prime exactly when $n = 1$ (`prime_positions_on_drift_axis`) — primes sit one successor from the basepoint and nowhere else; the inversion-dual position $1/(1/a+1/b) = ab/(a+b)$ (`inversion_dual_position`) applied to a prime puts $p/2$ at a non-integer position for every odd prime (`prime_inversion_misses_integers`); and an irrational drift step produces an axis with no integer points at all (`irrational_drift_axis_no_int_points`).

The six statements test one intuition: that prime-ness is a positional property. If primes are positions on an axis, then a change of drift step moves everything except the basepoint-adjacent slot, and the primality of $n \cdot p$ is destroyed exactly when $n > 1$. The inversion-dual statement is the first appearance of the $1/a + 1/b$ structure from the thinking document:

the symmetric position of a prime under the dual map is never a lattice position. The irrational-axis statement is the boundary case: a step that is not rational destroys the lattice entirely, so prime positions require the step to be an integer. All six are elementary facts about $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$; the proofs are direct, and each theorem's status in the ledger is CK.

5.2 The complex axis by projection (C011)

The projection construction is the base of the whole chain. The axis *ComplexAxis* is the set of pairs $\langle a, b\rangle$ with real a, b; $J = \langle 0, 1\rangle$ satisfies $J^2 = -1$ (`J_sq`), so $\sqrt{-1}$ exists in the axis (`sqrt_neg_one_exists_high`) while no real t satisfies $t^2 = -1$ (`sqrt_neg_one_not_exists_axis`). The projection proj is linear on the real component and annihilates J (`proj_add`, `proj_J`); it is a retraction of the embedding `lift` (`proj_lift`) but does not preserve products (`proj_mul_not_preserved`: $\mathrm{proj}(J \cdot J) \neq \mathrm{proj}\, J \cdot \mathrm{proj}\, J$). The basepoint projects to 0 (`basepoint_proj`) and the successor action projects to translation by 1 (`proj_succ`).

The key structural facts are three. First, basepoint drift is unobservable: every pure-imaginary basepoint e with $\mathrm{proj}\, e = 0$ produces the same projected chain as the default basepoint (`proj_chain_basepoint_independent`). Second, every axis line $\{z : z = \mathrm{lift}(t) + \mathrm{lift}(b) \cdot J\}$ projects onto the whole real axis (`axisLine_proj_eq_univ`) and the projected images of two different axis lines are identical (`axisLine_proj_independent`): the real axis is one axis of the same kind as any other, and under projection all axes are the same line (`realAxis_indistinguishable_from_line`). Third, recovery of the full structure from the projection is possible only up to the symmetry that projection destroys (`projection_recovery_theorem`).

The mathematics of this link is the elementary algebra of the complex numbers viewed as $\mathbb{R}^2$ with a preferred projection; every statement is a restatement of facts about $\mathbb{C}$ that mathlib proves, and the status is CK. What the formalization adds is the precise statement of the "origin illusion": no projected observation can distinguish the real axis from any other axis of the same kind, so any claim about the specialness of the real axis is not an observation but a choice. This matters in Section 6: the critical line is defined by a projected coordinate ($\mathrm{Re}(s) = 1/2$), and the map makes explicit that this coordinate is a projection choice, not an intrinsic marker.

5.3 The Riemann direction in the projection frame (C013)

The partial sums of the zeta series are constructed in the axis: `zetaTerm n s = lift (1/((n+1)^s))` (`zetaTerm_lift`), with the empty partial sum at `zetaPartial_zero` and the two-term case at `zeta_two_terms`. This link anchors the series that the Euler product below equals, and it fixes the notation used by the conditional restatement in C025. The construction is deliberately elementary: the partial sums are lifted terms of the axis, and their projection is the ordinary partial sum of the zeta series. The status is CK.

5.4 Prime reachability and prime circles (C014–C017)

Two drift steps around the unit circle send the integer point $n+J$ to the projected position $1+r^2 n/(n^2+1)$ (`doubleDrift_proj`). The formula is a direct computation: the second drift step applies the circle-inversion-like map of the axis to $n + J$ and projects. The projected image depends on n through the rational function $n/(n^2 + 1)$, which is bounded and small for large n — the double drift compresses the axis, and the compression is the first geometric expression of "the primes are reachable by drift".

A prime p is reachable: for every divisor m of $p - 1$ there is a drift that hits p (`prime_reachable_by_drift`). Primes congruent to 1 or 2 modulo 4 are norms of axis elements — Fermat's two-square theorem (`prime_two_axis`). The circle of radius $\sqrt{p}$ is the locus $\mathrm{norm}\, z = p$ (`primeCircle`); norm is multiplicative (`norm_mul`) and the integer lattice points of the circle close under the 90° rotation orbit (`orbit_closed`, `lattice_mul`). The two-square decomposition of a prime is unique — the uniqueness is the unique factorization of the Gaussian integers (`prime_sq_add_sq_unique`). The reflection $s \mapsto 1 - s$ arises as the square of a square-root map (`reflectSqrt`).

The links C014–C017 establish the bridge between the number-theoretic content of the direction (primes as positions, primes as norms) and the geometric content (circles, orbits, uniqueness by factorization). Each statement is classical; the formalization's contribution is the exact syntax of the bridge, and the status of each link is CK.

5.5 Critical-line geometry (C019–C022)

The central geometric fact of the chain: a point of the axis lies on the critical line exactly when it is its own conjugate under reflection, $\mathrm{proj}\, z = 1/2 \leftrightarrow 1 - z = \mathrm{conj}\, z$ (`critical_line_iff_conj` the critical line is parameterized as $\frac{1}{2} + it$ (`critical_line_points`). Under reciprocal (circle inversion), the critical line is exactly the circle centered at 1 with radius 1: $\|\mathrm{recip}(\frac{1}{2} + it) - 1\| = 1$ (`critical_line_is_circle`). This circle is the *critical circle* `criticalCircle` $= \{w : \mathrm{norm}(w - 1) = 1\}$, which passes through 0 and 2 (`zero_in_criticalCircle`); the real axis meets it exactly at those two points (`realAxis_inter_criticalCircle`) and the prime-2 circle meets it at $\langle 1, \pm 1\rangle$ (`prime2Circle_inter_critical`

The set of hypothetical nontrivial-zero positions, `nontrivialZeroSet` $= \{z : \mathrm{proj}\, z = 1/2 \wedge z.b \neq 0\}$, is carried by the reciprocal into the critical circle (`nontrivial_zero_on_circle`, `zeroSet_in_criticalCircle`) and, conversely, every point of the critical circle other than 0 and 2 is the reciprocal image of some such position (`criticalCircle_subset_zeroSet_image`). The two inclusions make the circle of zero positions and the critical circle the same object.

Two remarks keep the boundary honest. First, each of these statements is a pure geometric identity; none refers to the analytic value of ζ at any point, and the status of each link is CK. Second, the special points 0 and 2 have a geometric meaning: 0 is the image of the point at infinity under inversion (`recip_vanishes_at_infinity`) and 2 is the image of $\frac{1}{2}$ (`halfBasepoint_recip` and the real-axis intersection statement). The excluded points of `criticalCircle_subset_zeroSet_image` are exactly the images of the two points of the critical line that are not hypothetical zero positions.

5.6 Euler product and the zero-free region (C025)

For $\mathrm{Re}(s) > 1$ the Euler product equals the series and both equal $\zeta(s)$ (`zeta_euler_product`, `riemannZeta_euler_product`). mathlib's classical result gives $\zeta(s) \neq 0$ for $\mathrm{Re}(s) \geq 1$ (`riemannZeta_ne_zero_of_one_le_re`). The chain's last machine-checked statement is conditional: if $s \neq 0$, $\zeta(s) = 0$, and $\mathrm{Re}(s) = 1/2$, then $\|1/s - 1\| = 1$ (`zeta_zero_on_circle`). The hypothesis is an assumption of this statement, not a conclusion.

The Euler-product link is where the direction meets the analytic theory: the product over primes converges exactly where the series converges, and the zero-free region follows from the product's convergence. The conditional restatement is the only statement in the chain that mentions both ζ and the circle; it is deliberately conditional, and its role in the map is to show that the geometry and the analysis are compatible: a zero on the critical line is a point of the critical circle. The status of this link is CK.

6 The gap

The verification map has one open edge. The machine-checked chain proves three things: (i) the critical line is the circle $\{w : \|w - 1\| = 1\}$ under inversion (C019–C022); (ii) the hypothetical zero positions sit on that circle (C021–C022); (iii) the Euler product converges exactly on $\mathrm{Re}(s) > 1$ and ζ has no zeros on $\mathrm{Re}(s) \geq 1$ (C025). It does not prove the remaining statement: that every nontrivial zero of ζ lies on the critical line.

To close the gap a proof would have to show that no zero of ζ exists off the line $\mathrm{Re}(s) = 1/2$ within the strip $0 < \mathrm{Re}(s) < 1$. The geometry of this manuscript neither assumes nor implies that statement. Two structural remarks make the gap precise.

First, the gap is not hidden inside the chain. No edge of the dependency graph of Section 1.4 is unlabeled, and the only CONJ node is the final edge — the map's purpose is to make exactly this edge visible to the referee.

Second, the gap has a precise geometric shape. By C019–C022, a hypothetical zero off the critical line would project to a point of the axis with $\mathrm{proj}\, z \neq 1/2$, and its reciprocal would lie on a circle other than the critical circle. Within this map, the question "does ζ have a zero off the line?" is equivalent to the question "does the reciprocal of a zero of ζ leave the critical circle?". The map reduces the analytic question to a geometric one — but it does not answer it. Any future proof that closes the gap must show, by analytic means outside the current chain, that the reciprocal of every nontrivial zero stays on the critical circle.

7 Numerical evidence

The visualization of 2026-08-06 produced 138,067 zeros on the critical line, computed with the Riemann–Siegel formula (bases 2–36 and Gaussian-integer complex bases; the critical-line-only layout followed at 17:59–18:09; the persisted prime/zero data at 18:35–18:46). The zero-location error was reduced from 0.1 to 9×10^{-4} by a correction to the Riemann–Siegel formula. This is observational evidence, not proof; it is recorded because it is the trail's starting point and because the verification map separates it cleanly from the machine-checked chain: the paragraph you are reading is the only place in this manuscript where numbers computed rather than proved appear. External numerical verification of the first 10^{13} zeros (real parts equal to 1/2) is likewise observational and is not part of any machine-checked statement.

8 Honest boundary

Every claim in this manuscript is labeled `novelty_status: KNOWN`: each is a restatement of classical mathematics (circle inversion, Gaussian unique factorization, Fermat's two-square theorem, the Euler product, elementary projection algebra) with a machine-checked proof. No claim is made of proving the Riemann hypothesis; mathlib's `RiemannHypothesis` remains unproved. The numerical evidence is observational. The value of the map is that it separates, with mechanical precision, what is proved from what is conjectured.

This boundary is not an apology; it is the method. The credibility of the manuscript rests not on the strength of its claims but on the exactness of their labeling. A referee who disagrees with any label can verify it in minutes; a referee who agrees with all labels has a complete picture of what is proved, what is classical, and what is open. The boundary applies to every paragraph of this manuscript: each paragraph states only what its node of the map supports, and nothing more.

9 Related work

The formalization of the Riemann zeta function in mathlib (as an L-series with meromorphic continuation) provides the analytic base of the chain [1, 4]. Classical references for the statement, the functional equation, and the Euler product are [2, 3, 5]. The verification-map form follows the general program of machine-checked mathematics: the standard of "the manuscript carries its own proof of correctness" is the standard of interactive theorem proving, applied here to the structure of a submission rather than to its individual steps. The novelty of the present manuscript is not mathematical — every machine-checked statement is known — but organizational: a template for independent researchers in which the manuscript's trustworthiness is decidable from the manuscript itself.

10 Conclusion

A manuscript organized as a verification map makes its own trust decision mechanical. Fifteen links of the Riemann-direction chain are machine-checked against Lean 4.32.2 / mathlib v4.32.2 with zero `sorry` and only the standard axioms; the sixteenth edge — the Riemann hypothesis itself — is stated exactly, named exactly, and left open. A referee's question is reduced to: is each node of the map what it claims to be? The answer to that question is a file check, not an act of faith.

Appendix: theorem inventory

The full inventory of the machine-checked chain, by file. All theorems compile with zero errors and zero `sorry`.

`PrimeDriftPositions.lean` ***(6 theorems).*** `drift_vector_is_successor`, `successor_positions_in_drift_frame`, `prime_positions_on_drift_axis`, `inversion_dual_position`, `prime_inversion_misses_integers`, `irrational_drift_axis_no_int_points`.

`ComplexAxis.lean` ***(128 theorems; selection).*** Structure: `ext`, `J_sq`, `proj_add`, `proj_J`, `proj_lift`, `lift_add`, `lift_mul`, `proj_mul_not_preserved`, `sqrt_neg_one_exists_high`, `sqrt_neg_one_not_exists_axis`, `lift_neg_one_sqrt`. Basepoint: `basepoint_proj`, `pure_imag_proj`, `proj_succ`, `zero_in_proj_chain`, `proj_chain_succ_closed`, `proj_chain_basepoint_independent`. Axis lines: `axisLine_eval_proj`, `axisLine_proj_all`, `axisLine_proj_injective`, `axisLine_proj_eq_univ`, `axisLine_proj_independent`, `realAxis_indistinguishable_from_i_line`. Inversion: `recip_mul_self`, `recip_lift`, `proj_recip_lost`, `circleInv_core`, `circleInv_fixed`, `circleInv_proj_preserved`, `circleInv_translate`, `norm_recip`, `recip_vanishes_at_infinity`, `circleInv_vanishes_at_infinity`. Partial sums: `zetaTerm_lift`, `zetaPartial_zero`, `zeta_two_terms`. Critical line: `one_minus_fixed`, `one_minus_fixed_cplx`, `critical_line_iff_conj`, `reflectSqrt_sq`, `reflectSqrt'_sq`, `critical_line_is_circle`, `critical_line_points`, `real_axis_points`, `nontrivial_zero_position`. Drift reachability: `doubleDrift_proj`, `prime_reachable_by_drift`, `prime_two_axis`. Circles: `norm_mul`, `lattice_mul`, `orbit_closed`, `prime_sq_add_sq_unique`, `primeCircle`, `criticalCircle`, `zero_in_criticalCircle`, `primeCircle_recip`, `critical_line_in_circle`, `realAxis_inter_criticalCircle`, `inter_proj`, `prime2Circle_inter_criticalCircle`, `halfCircle_recip`, `nontrivial_zero_on_circle`, `critical_line_image_subset_circle`, `nontrivialZeroSet`, `zeroSet_in_criticalCircle`, `criticalCircle_subset_zeroSet_image`. Products and splitting: `mul_conj`, `prime_conj_pair`, `rotated_conj_pair`, `J_pow_two`, `J_pow_four`, `rotate_two_neg`, `recip_axis_second_order`, `zero_point_prime_pair`, `norm_unit4`, `associates_norm`, `variants_are_associates`, `conj_recip`, `recip_conj_pair`, `critical_line_circle_on_recip_axis`. Viewpoint: `halfBasepoint_proj`, `halfBasepoint_on_critical_line`, `basepoint_is_critical_line_family`, `basepoint_drift_to_half`, `halfBasepoint_in_realAxisAtHalf`, `realAxisAtHalf_proj`, `halfBasepoint_recip`, `realAxisAtHalf_recip`, `proj_kernel_J`, `lift_injective`, `real_axis_preserved_by_proj`, `proj_not_recoverable`, `proj_recoverable_symmetry`, `projection_re`, `primes_countable`, `pureImag_collapse`, `basepoint_ambiguity`, `collapse_kernel`.

`ZetaEulerProduct.lean` ***(7 theorems).*** `zetaEulerF`, `zetaEulerF_no`, `zeta_euler_product`, `riemannZeta_euler_product`, `riemannZeta_ne_`, `critical_line_iff_conj_complex`, `verified_zero_on_circle`, `zeta_zero_on_circle`.

`Toolkit/CriticalPrimeCircles.lean` ***(R145, 5 theorems).*** `critical_circle_points`, `prime_circle_center_zero`, `reciprocal_p`, `log_pair_instance`, `fold_centers_are_circle_centers`.

`Toolkit/PatRiemannTwinPrimes.lean` ***(7 theorems).*** `critical_lin`, `zero_on_pat_circle` (conditional), `prime_on_pat_circle`, `prime_log_monophase`, `twin_gap_is_circle_diameter`, `twin_gap_inv`, `pat_riemann_twin_perspective`.

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